

Train, Validation, Test

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can select a model using a declared validation rule.

I can keep training, validation and test roles separate.

I can recognize repeated validation as development.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Hold Out Test Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Absolute Error and MAE. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Training fits parameters, validation chooses settings, and testing estimates final performance after choices are frozen.



LEARNING TARGET You have 30 leaves with 3 views each. Allocate 18 leaves to train, 6 to validation, 6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Builder name and experiment date:

Readiness for the connected explanation

Which data fitted the line last week?

This week's end goal

By the end, I can explain train, validation, test using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use train, validation, test to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

Fitting and selecting answer different questions. Training adjusts parameters. Validation helps choose settings or model families. A final test estimates the performance of the frozen development process on held-out examples. Keep related records in the same split when separate rows could reveal the same source.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason. Earlier weeks supplied the model-building process; this week examines validation and model selection.

3. See how it works

ML MECHANISM

Input → Model → Evidence



IN

INPUT

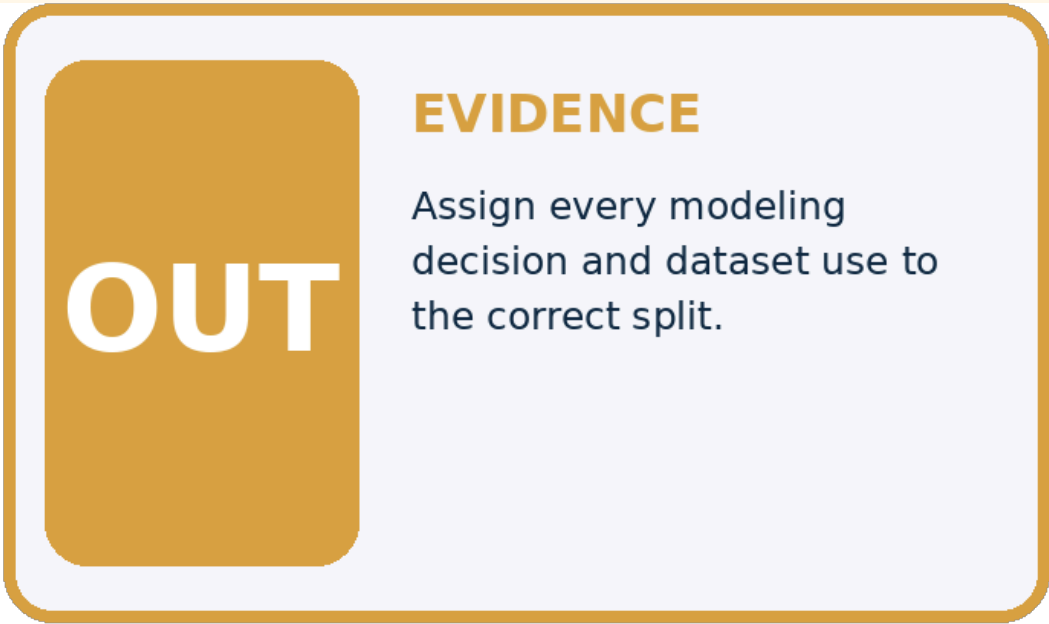
Assign every modeling decision and dataset use to the correct split.



ML

MODEL

Training fits parameters, validation chooses settings, and testing estimates final performance after choices are frozen.



OUT

EVIDENCE

Assign every modeling decision and dataset use to the correct split.

QUESTION Assign every modeling decision and dataset use to the correct split.

MODEL IDEA Training fits parameters, validation chooses settings, and testing estimates final performance after choices are frozen.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Fit two candidate trees on training rows. Choose depth and threshold using validation metrics and error slices.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

Three candidates have training MAE $A=1$, $B=0$, $C=2$ and validation MAE $A=3$, $B=5$, $C=3$. The declared rule is smallest validation MAE, then simpler model, with A simpler than C. Choose A. B fits training perfectly but does worse on validation. The final test remains unopened.

Before continuing, underline the decision rule and circle the evidence that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim. Keep the distinction between an observed result and a proposed next step.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: A and B both have validation MAE2 cm; A is simpler under a predeclared tie rule.

Prediction: Select A using the predeclared simpler-model tie-break.

Reason: The validation errors tie, so the stated second arbitrary rule decides without consulting final data.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

VALIDATION SET

data used to choose settings

3

TEST SET

data used for final estimate

4

SPLIT STRATEGY

rule assigning examples to sets

Term · Precise meaning · My ML example

Term: training set

Precise meaning: data used to fit parameters

My ML example: _____

Term: validation set

Precise meaning: data used to choose settings

My ML example: _____

Term: test set

Precise meaning: data used for final estimate

My ML example: _____

Term: split strategy

Precise meaning: rule assigning examples to sets

My ML example: _____

MISCONCEPTION TO REPAIR Three random folders are enough even when photos of one plant appear in each.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Validation

Meaning: Development evidence used to choose between options

Word: Hyperparameter

Meaning: A model setting chosen outside its fitted parameters

Word: Selection rule

Meaning: A declared method for choosing an option

Word: Test set

Meaning: Held-out evidence reserved for a final evaluation

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 24 • WORKED EVIDENCE

- 1 Fit two candidate trees on training rows.
- 2 Choose depth and threshold using validation metrics and error slices.
- 3 Freeze preprocessing, features, model, and decision threshold.
- 4 Evaluate once on test; do not return to tune and still call it final.

STEP 1 Fit two candidate trees on training rows.

STEP 2 Choose depth and threshold using validation metrics and error slices.

STEP 3 Freeze preprocessing, features, model, and decision threshold.

STEP 4 Evaluate once on test; do not return to tune and still call it final.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

Validation and model selection at a glance

Part · Evidence or action · Interpretation

Part: Training

Evidence or action: Fit each candidate

Interpretation: No test labels

Part: Validation

Evidence or action: Choose A by rule

Interpretation: Record every attempt

Part: Final test

Evidence or action: Frozen model only

Interpretation: Report after selection

A tie policy prevents a convenient choice after inspecting results. Equal validation scores do not prove equal future performance. Record candidate settings, split identifiers, preprocessing and the full comparison, including rejected options.

Where should parameter fitting, setting selection and final reporting occur? Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Train, Validation, Test

Assign a separate job to training, validation and final test evidence.

The facts we will use

Candidates A and B have training MAE1 and 0 cm; validation MAE2 and 3 cm. Selection rule is smallest validation MAE.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Train, Validation, Test

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit on training

Both candidates are built using training data; B fits it more closely.



2. Select on validation

Choose A because 2 cm is below 3 cm.



3. Reserve final test

Freeze A before scoring final cases; do not select again using their labels.

CHANGE ONE THING

A and B both have validation MAE 2 cm; A is simpler under a predeclared tie rule.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Fit on training

What happens and why: Both candidates are built using training data; B fits it more closely.

Step: 2. Select on validation

What happens and why: Choose A because 2 cm is below 3 cm.

Step: 3. Reserve final test

What happens and why: Freeze A before scoring final cases; do not select again using their labels.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Freeze A before scoring final cases; do not select again using their labels. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Repeated validation use can itself overfit model choices; keep a genuinely untouched final evaluation.

Read the labelled picture: Why does B's zero training MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: A and B both have validation MAE2 cm; A is simpler under a predeclared tie rule. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: Three random folders are enough even when photos of one plant appear in each. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE Three random folders are enough even when photos of one plant appear in each.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A low score is meaningful only with its unit, metric, evaluated population and baseline. Decide how to handle ties and missing predictions before comparison. If no appropriate final test exists, state that limitation and keep claims about validation.

Why is the winning validation score not a guaranteed future score?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

If you inspect validation errors and invent candidate D, that is another development attempt. Log it. Repeated experiments can adapt to peculiarities of validation. Do not rename validation as an untouched test. More rows alone do not repair duplicated groups or future information leaking into inputs.

Practise with less help: Assign data roles

Use workbook W2 for the connected example “Validation and model selection”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Validation and model selection. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Why does B’s zero training MAE not settle the original choice? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Role Assignment: sort twenty actions into train/validation/test, then group related examples before splitting.

1. Write the hypothesis and expected evidence before running.
2. Role Assignment: sort twenty actions into train/validation/test, then group related examples before splitting.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

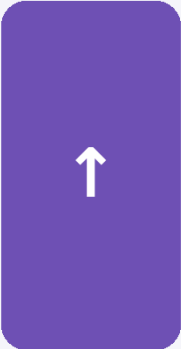
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: training set, validation set, test set, split strategy.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE Three random folders are enough even when photos of one plant appear in each.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.



RESPONSIBLE PLANT CLASSIFIER
Publish the grouped split strategy and counts for the Plant Classifier, including why plant identity stays within one split.

THIS WEEK'S CHECKPOINT Publish the grouped split strategy and counts for the Plant Classifier, including why plant identity stays within one split.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Assign every modeling decision and dataset use to the correct split.

REQUIRED PROOF Assign every modeling decision and dataset use to the correct split.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Metrics Match Consequences.

13. Study a complete case

Week 24 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Split 100 independent leaves into 60 train,20 validation,20 test.

The question

Explain roles and preprocessing.

Worked explanation

Train fits parameters and learned transforms; validation chooses settings; final test evaluates the selected system. Group related records first.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Three partitions have three different jobs

Evidence or stage · Value or result

Evidence or stage: Train, fit parameters

Value or result: 60

Evidence or stage: Validation, choose settings

Value or result: 20

Evidence or stage: Test, evaluate selected system

Value or result: 20

Split 100 independent leaves into 60 train,20 validation,20 test.

Explain roles and preprocessing.

Train fits parameters and learned transforms; validation chooses settings; final test evaluates the selected system. Group related records first.

14. Try a new case independently

Independent case: Apply independently

Training errors P 2 Q 0 R 1; validation errors P 4 Q 6 R 4. Rule: smallest validation MAE, then simpler; P is simpler than R. Choose and state the test action.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 24 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Assign roles: set A fits model parameters; B selects between two models; C is opened once after choices are fixed. Which is train, validation and test?

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Publish the grouped split strategy and counts for the Plant Classifier, including why plant identity stays within one split.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

You have 30 leaves with 3 views each. Allocate 18 leaves to train,6 to validation,6 to test. How many images are in each? Where is a learned normalisation mean fitted?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Assign a separate job to training, validation and final test evidence.

By the end, I can · How I will show it

By the end, I can: Select a model using a declared validation rule

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Keep training, validation and test roles separate

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Recognize repeated validation as development

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Select a model using a declared validation rule

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.